

Prime Factorization

Lesson 1-1

Name: _____

Date: _____

Class: _____

Key Vocabulary

Level 2 Standard

Picture first, then the word, then a plain-language meaning. Say each word out loud.



factors: 1 and 7

7 has only two factors: 1×7 . So 7 is prime.

Prime number

Write the definition:



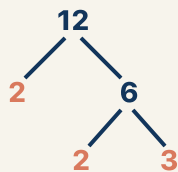
$6 = 1, 2, 3, 6$

many factors $\rightarrow$ composite

$12 = 1 \times 12, 2 \times 6, 3 \times 4$ — six factors, so 12 is composite

Composite number

Write the definition:

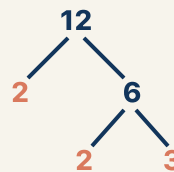


$$12 = 2 \times 2 \times 3$$

$$36 = 2 \times 2 \times 3 \times 3 = 2^2 \times 3^2$$

Prime factorization

Write the definition:



$$12 = 2 \times 2 \times 3$$

$$24 \rightarrow 4 \times 6 \rightarrow (2 \times 2) \times (2 \times 3) \rightarrow 2 \times 2 \times 2 \times 3$$

Factor tree

Write the definition:

$$5^3$$

$$5 \times 5 \times 5$$

2^3 means $2 \times 2 \times 2 = 8$

Exponent

Write the definition:

Guided Notes

Level 2 Standard



WHAT WE'RE LEARNING TODAY

I can write a number as a product of its prime factors using a factor tree.



Fill in each blank as we go. Use the Word Bank to help you.



WORD BANK — FILL EACH BLANK WITH THE BEST WORD

Prime number

Composite number

Prime factorization

Factor tree

Exponent

1

A whole number greater than 1 with exactly two factors, 1 and itself, is a _____ number.

2

A whole number greater than 1 that has more than two factors is a _____ number.

3

_____ — Writing a number as prime numbers multiplied together.

4

I can break a number into its prime factors step by step using a _____.

5

In 2^3 , the small number 3 is the _____ and it shows 2 is multiplied 3 times.



Watch & Try — Worked Examples

See the notes in action: watch one worked all the way through, then try the next with the same steps.

 I do — watch

Follow each step as your teacher solves it.

Problem: Which of the following is a prime number?

- A. 17
- B. 15
- C. 21
- D. 9

Step 1 17 has exactly two factors: 1 and 17.

Step 2 $15 = 3 \times 5$, $21 = 3 \times 7$, and $9 = 3 \times 3$, so they are all composite.

 **Answer:** A. 17

 We do — together

Solve this one with your class using the same steps.

Problem: What is the prime factorization of 30?

- A. $2 \times 3 \times 5$
- B. 5×6
- C. 2×15
- D. 3×10

Step 1 _____

Step 2 _____

Answer: _____

 You do — your turn

Now try one on your own. Show every step.

Problem: What is the prime factorization of 18?

- A. $2 \times 3 \times 3$
- B. 2×9
- C. 3×6
- D. 6×3

Show your work:

Try It

Solve on your own. Check the answer key when you are done.

1. The second panel needs a number whose prime factorization is $2^2 \times 3^2$. Which number unlocks it?

- A. 36
- B. 24
- C. 12
- D. 18

Show your work:

2. The final lock asks for the prime factorization of 78 written as a product of primes. Which is correct?

- A. $2 \times 3 \times 13$
- B. 2×39
- C. 3×26
- D. 6×13

Show your work:

Stretch Your Thinking

Level 2 enrichment

Challenge task — explain your reasoning in full sentences.

Choose any two-digit composite number. Show TWO different factor trees that both lead to the same prime factorization. Explain why every composite number has only one prime factorization.

Sentence starter: I chose the number _____. My first factor tree starts with _____ $\times$ _____, and my second starts with _____ $\times$ _____. Both give the same prime factorization: _____. This happens because _____.

Show your work:

Reflect — Exit Ticket

What is the prime factorization of 40?

- A. $2 \times 2 \times 2 \times 5$
- B. 4×10
- C. 5×8
- D. 2×20

Your answer:

Answer Key & Teacher Guide

1. **Notes 1:** Prime number
2. **Notes 2:** Composite number
3. **Notes 3:** Prime factorization
4. **Notes 4:** Factor tree
5. **Notes 5:** Exponent
6. **We Try (notes):** A. $2 \times 3 \times 5 = 30 = 2 \times 15 = 2 \times 3 \times 5$. All three factors (2, 3, 5) are prime, so $2 \times 3 \times 5$ is the prime factorization.
7. **You Try (notes):** A. $2 \times 3 \times 3 = 18 = 2 \times 9 = 2 \times 3 \times 3$. Both 2 and 3 are prime, so $2 \times 3 \times 3$ is the prime factorization.
8. **Try It 1:** A. $36 = 2^2 \times 3^2 = 4 \times 9 = 36$, so 36 is the code for this panel.
9. **Try It 2:** A. $2 \times 3 \times 13 = 78 = 2 \times 39 = 2 \times 3 \times 13$. All three factors are prime, so $2 \times 3 \times 13$ unlocks the final panel.
10. **Exit Ticket:** A. $2 \times 2 \times 2 \times 5 = 40 = 2 \times 20 = 2 \times 2 \times 10 = 2 \times 2 \times 2 \times 5$. All factors (2, 2, 2, 5) are prime.